



American International University-Bangladesh (AIUB)

Department of Computer Science
Faculty of Science & Technology (FST)

PROJECT REPORT

Rural, Underwater & Titanic Scenarios: 2D Animation System Using OpenGL (GLUT)

A Computer Graphics Project Submitted

By

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1. Introduction

This project presents an interactive multi-scenario 2D animation system built using OpenGL and the OpenGL Utility Toolkit (GLUT) in C++. The system integrates three independent animated environments a rural transportation scene, an underwater exploration scene, and a Titanic ship-collision scene into a single executable application.

Users switch between the three scenarios in real time using the keyboard (keys 1, 2 and 3) and can trigger additional interactions within the underwater scenario using dedicated keys. The project uses immediate-mode OpenGL rendering to draw all objects.

The work includes 2D scene design, procedural (sine-based) motion, alpha-blending for transparency, coordinate transformations, and keyboard-driven interaction all being implemented without the use of a game engine, which keeps the fundamental graphics pipeline completely visible and under control.

2. Project Objectives

The project's main objective is to build a working multi-scenario 2D animation system in OpenGL/GLUT. The specific objectives are:

1. Design and render three distinct 2D animated scenarios (rural, underwater and ship-collision) using OpenGL/GLUT primitives.
2. Implement a keyboard-driven scenario-switching system that lets the user move between scenes instantly using the number keys 1, 2 and 3.
3. Apply procedural, time-based animation (trigonometric motion, alpha blending and randomized elements) to simulate realistic environmental motion such as waves, swaying grass and drifting bubbles.
4. Demonstrate transformation concepts (translation and rotation) through interactive elements such as the anchor-lifting mechanism and the sinking/rotating Titanic ship.

3. Project Overview and Features

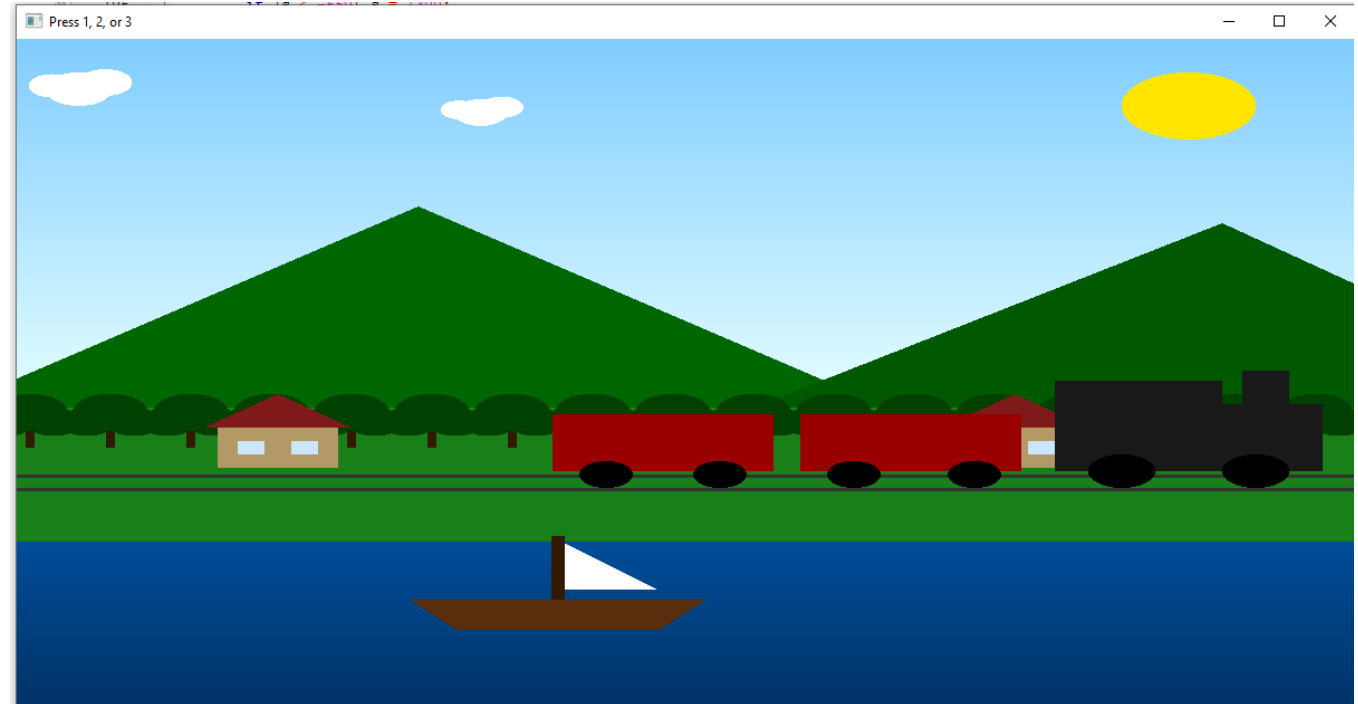
The application opens with the rural scene by default. Pressing 1, 2 or 3 instantly switches the active scenario without restarting the program, and the underwater scenario additionally responds to P (start marine life movement) and L (lift anchor). The major features are summarized below.

SL	Feature	Brief Description
1	Multi-scenario management	A single global currentScenario variable controls which of the three scenes is drawn and updated each frame.
2	Real-time animation loop	A GLUT timer callback (update()) advances all animation variables roughly every 16 ms.
3	Keyboard-based interaction	Keys 1/2/3 switch scenarios; P and L trigger start and anchor-lift actions in Scenario 2.
4	Independent projection systems	Each scenario sets its own gluOrtho2D() range, so coordinate systems do not need to match across scenes.
5	Procedural animation & blending	Sine-based motion for waves, grass and jellyfish, randomized bubbles, stars and alpha blending for transparency.

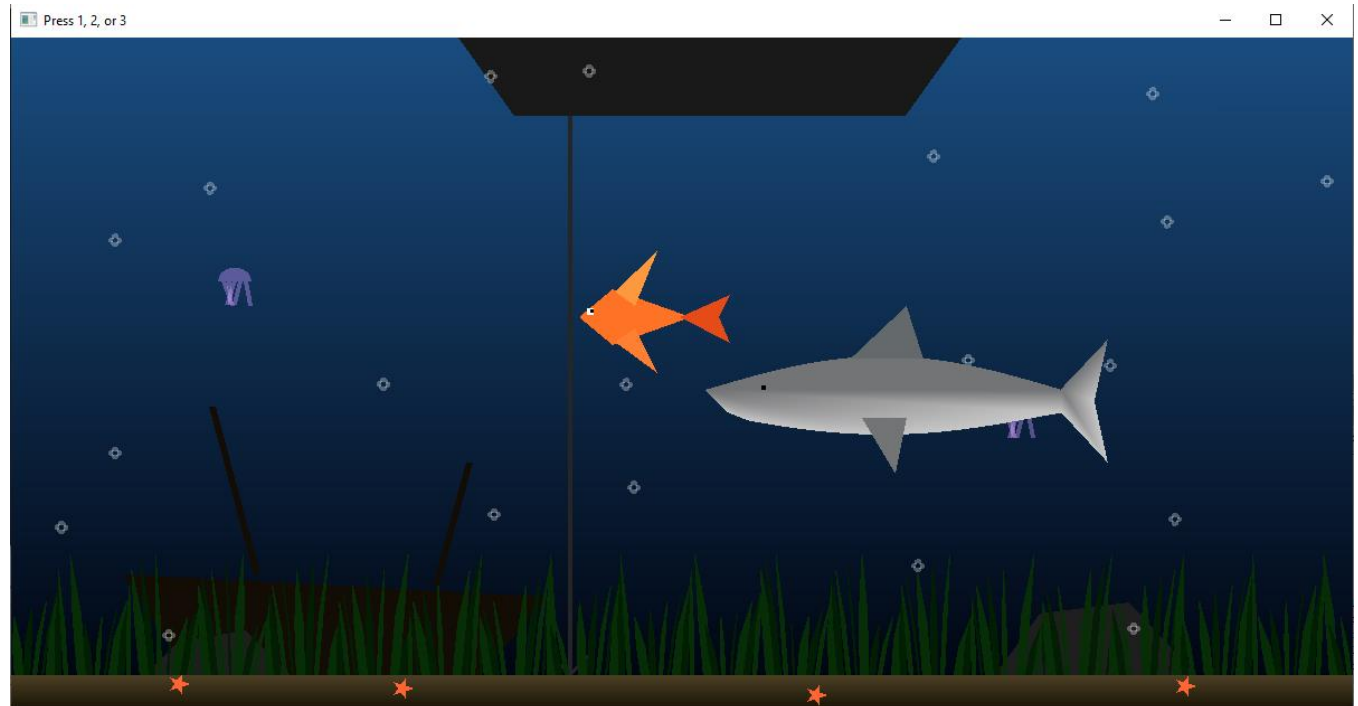
4. Project Graph / Scene Design

Scenario 1 (Rural Transportation) and Scenario 3 (Titanic Collision) both use a normalized orthographic projection of gluOrtho2D(-1, 1, -1, 1), with the origin (0,0) at the centre of the screen. Scenario 2 (Underwater Exploration) instead uses a pixel-based projection of gluOrtho2D(0, 1200, 0, 600), matching the 1200x600 window, with the origin at the bottom-left corner. The screenshots below show each scenario's layout.

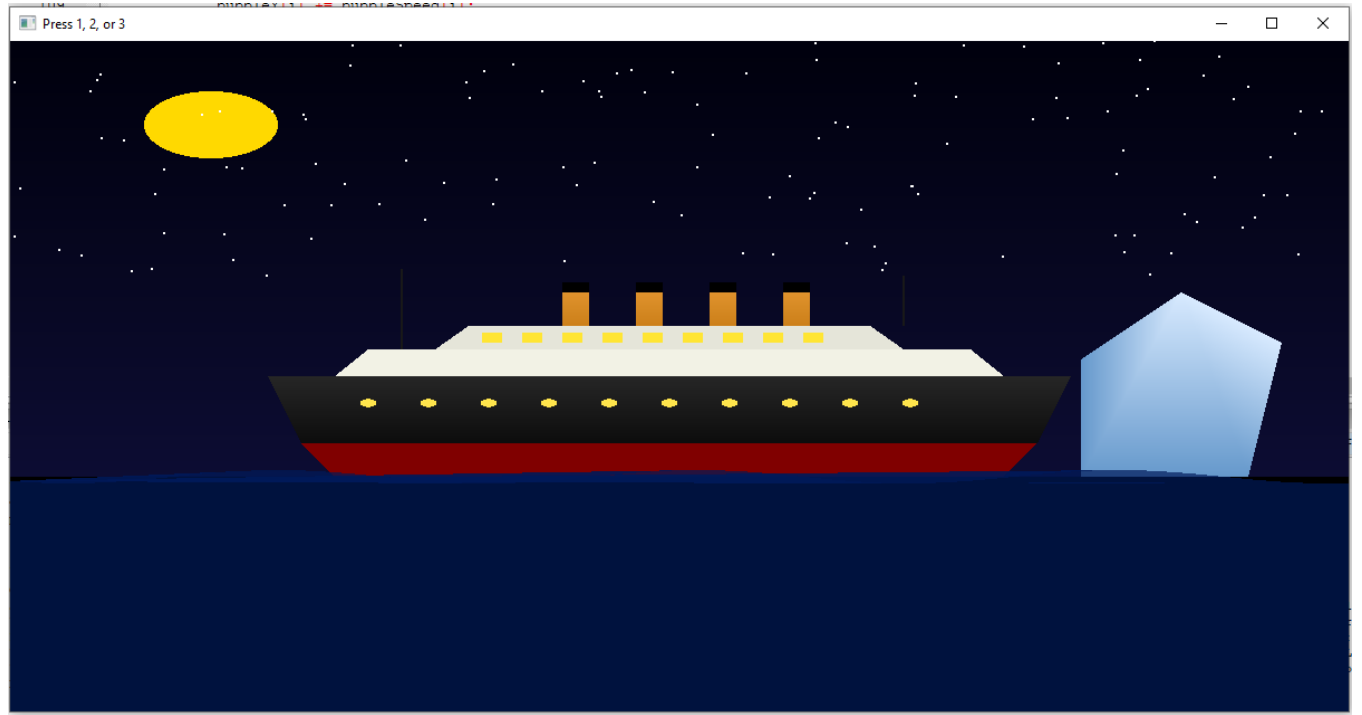
Scenario 1 (Rural Transportation)



Scenario 2 (Underwater Exploration)



Scenario 3 (Titanic Collision)



4.1 Coordinate / Transformation Notes

Scenario 1: groundLevel (-0.34) marks the ground line. The train, boat, clouds, and birds are moved with `glTranslatef()` and simple incremental offsets, wrapping back to the opposite edge once they leave the -1 to 1 view.

Scenario 2: all objects live in the 0 to 1200 by 0 to 600 pixel space. The surface ship and fish use additive positional offsets; the anchor uses `glTranslatef()` and a vertical offset (`anchorY`) that increases when lifting, which then triggers the ship's horizontal motion.

Scenario 3: the Titanic is moved with `glTranslatef(tShipX, tShipY - sink, 0)` combined with `glRotatef(-tAngle, 0, 0, 1)` to simulate sinking (vertical translation) and rotation after the iceberg collision.

5. List of Objects and Object IDs

SL#	Object ID	Object Name	Description / Role
1	OBJ-01	Train	Rural scene: moving locomotive and carriages travelling along the track
2	OBJ-02	River Boat	Rural scene: sailboat drifting on the river with a gentle bobbing motion
3	OBJ-03	Clouds	Rural scene: drifting background clouds
4	OBJ-04	Birds	Rural scene: birds flying across the sky

SL#	Object ID	Object Name	Description / Role
5	OBJ-05	Trees	Rural scene: trees lining the riverbank
6	OBJ-06	Houses	Rural scene: houses along the landscape
7	OBJ-07	Fish1 (Orange Fish)	Underwater scene: orange fish swimming across the screen
8	OBJ-08	Fish2 (Red Fish)	Underwater scene: red fish swimming in the opposite direction
9	OBJ-09	Shark	Underwater scene: shark model built from a triangle fan and polygons
10	OBJ-10	Jellyfish	Underwater: pulsating jellyfish and waving tentacles
11	OBJ-11	Starfish	Underwater scene: starfish resting on the seabed
12	OBJ-12	Surface Ship & Anchor	Underwater scene: ship with a liftable anchor mechanism
13	OBJ-13	Sunken Ship	Underwater scene: shipwreck resting on the seabed
14	OBJ-14	Seabed Grass	Underwater scene: swaying sea-grass covering the seabed
15	OBJ-15	Rocks	Underwater scene: rock clusters on the seabed
16	OBJ-16	Bubbles	Underwater scene: rising bubbles drifting toward the surface
17	OBJ-17	Titanic Ship	Scenario 3: the ship that collides with the iceberg and sinks
18	OBJ-18	Iceberg	Scenario 3: the stationary iceberg the ship collides with
19	OBJ-19	Stars	Scenario 3: randomly placed background stars in the night sky

SL#	Object ID	Object Name	Description / Role
20	OBJ-20	Sea Surface	Scenario 3: animated, alpha-blended sea surface with waves
21	OBJ-21	Sun	Rural scene: yellow sun in the sky, drawn with the shared circle() helper
22	OBJ-22	Moon	Scenario 3: pale moon in the night sky, drawn with the shared circle() helper
23	OBJ-23	Sky	Rural scene: blue-to-white gradient sky background behind the sun, clouds and birds
24	OBJ-24	Mountains	Rural scene: two triangular mountain ranges in the background landscape
25	OBJ-25	River	Rural scene: blue gradient river band at the base of the scene, on which the boat sails
26	OBJ-26	Water Background	Underwater scene: blue-to-dark gradient background representing the ocean water column
27	OBJ-27	Night Sky	Scenario 3: dark navy gradient background representing the night sky above the sea

6. Functions Used to Represent Objects

SL#	Object ID	Object Name	Function Name	Purpose
1	OBJ-01	Train	drawTrain()	Draws the locomotive body, cab, chimney, and wheels
2	OBJ-02	River Boat	drawBoat()	Draws the hull and sail, with a bobbing offset based on riverShift
3	OBJ-03	Clouds	drawCloud()	Draws layered circular cloud shapes
4	OBJ-04	Birds	drawBird()	Draws simple V-shaped birds in flight

SL#	Object ID	Object Name	Function Name	Purpose
5	OBJ-05	Trees	drawTree()	Draws the trunk and canopy using circles
6	OBJ-06	Houses	drawHouse()	Draws the house body, windows, and roof
7	OBJ-07	Fish1	fish1()	Draws the orange fish using colored polygons
8	OBJ-08	Fish2	fish2()	Draws the red fish using colored polygons
9	OBJ-09	Shark	Shark()	Draws the shark body, fin, and tail using a triangle fan
10	OBJ-10	Jellyfish	drawJellyfish()	Draws the pulsing bell and tentacles
11	OBJ-11	Starfish	drawStarfish()	Draws the five-pointed starfish shape
12	OBJ-12	Surface Ship & Anchor	drawSurfaceShip()	Draws the ship hull and the anchor/chain
13	OBJ-13	Sunken Ship	drawSunkenShip()	Draws the wrecked ship on the seabed
14	OBJ-14	Seabed Grass	drawOrganicGrass()	Draws the swaying grass blades
15	OBJ-15	Rocks	drawRocks()	Draws the rock clusters on the seabed
16	OBJ-16	Bubbles	drawBubbles()	Draws the rising circular bubble outlines
17	OBJ-17	Titanic Ship	ship()	Draws the hull, deck, and chimneys of the Titanic
18	OBJ-18	Iceberg	iceberg()	Draws the iceberg polygon
19	OBJ-19	Stars	stars()	Draws the randomly placed background stars
20	OBJ-20	Sea Surface	sea()	Draws the animated, alpha-blended sea
21	OBJ-21	Sun	drawSun()	Draws the sun as a filled circle (own

SL#	Object ID	Object Name	Function Name	Purpose
				function, calls the shared circle() helper)
22	OBJ-22	Moon	moon()	Draws the moon as a filled circle (own function, calls the shared circle() helper)
23	OBJ-23	Sky	drawSky()	Draws the blue-to-white gradient sky background
24	OBJ-24	Mountains	drawMountains()	Draws the two triangular mountain shapes
25	OBJ-25	River	drawRiver()	Draws the gradient river band at the bottom of the scene
26	OBJ-26	Water Background	waterBackground()	Draws the underwater gradient background
27	OBJ-27	Night Sky	nightSky()	Draws the dark gradient night-sky background

7. Animation and Interaction Functions

SL#	Function ID	Function Name	Object / Scene	Trigger	Brief Description
1	FN-01	display()	All scenarios	Every frame (GLUT)	Clears the screen and calls the active scenario's run function
2	FN-02	keyboard()	All scenarios	Key press (1,2,3,P,L)	Switches the active scenario; calls startScenario2() / liftScenario2Anchor() for Scenario 2's 'P'/'L' keys
3	FN-03	update()	All scenarios	GLUT timer (~16 ms)	Calls whichever updateScenarioN() matches the active scenario, then reschedules itself
4	FN-04	runScenario1()	Scenario 1	Called from display()	Sets the -1..1 projection and draws the rural scene
5	FN-05	runScenario2()	Scenario 2	Called from display()	Sets the 0..1200/0..600 projection and draws the underwater scene

SL#	Function ID	Function Name	Object / Scene	Trigger	Brief Description
6	FN-06	runScenario3()	Scenario 3	Called from display()	Sets the -1..1 projection and draws the collision scene
7	FN-07	Anchor-lift logic (in updateScenario2())	Scenario 2	Automatic once liftingAnchor is set (via 'L' key)	Raises anchorY until 530, then sets shipMoving to start the ship
8	FN-08	Sinking/rotation logic (in updateScenario3())	Scenario 3	Automatic after collision	Increases sink and tAngle each frame to sink and tilt the ship
9	FN-09	Train/landscape motion (in updateScenario1())	Scenario 1	Automatic every frame	Moves the train, boat, clouds, and birds, wrapping each back off-screen
10	FN-10	Fish & shark motion (in updateScenario2())	Scenario 2	Automatic once 'P' pressed	Moves fish1, fish2, and the shark, wrapping them off-screen
11	FN-11	Bubble animation (in updateScenario2())	Scenario 2	Automatic once 'P' pressed	Rises each bubble and resets it to the seabed once it reaches the surface
12	FN-12	Grass sway animation (in updateScenario2())	Scenario 2	Automatic once 'P' pressed	Increments grassSway to animate the swaying grass and jellyfish
13	FN-13	Ship approach & collision check (in updateScenario3())	Scenario 3	Automatic every frame	Moves the Titanic toward the iceberg until it collides (sets hit = true)
14	FN-14	Wave animation (in updateScenario3())	Scenario 3	Automatic every frame	Increments wave to animate the sea surface
15	FN-15	startScenario2()	Scenario 2	'P' key	Sets flag3S2 and movingS2 to begin Scenario 2's fish/shark/bubble/grass animation
16	FN-16	liftScenario2Anchor()	Scenario 2	'L' key	Sets liftingAnchor so updateScenario2() begins raising the anchor each frame
17	FN-17	initScenario2()	Scenario 2	Called once from main()	Seeds each bubble's starting position and rise speed
18	FN-18	initScenario3()	Scenario 3	Called once from main()	Seeds each star's random position in the night sky

8. Work Distribution / Contribution

SL	Member Name	Student ID	Implemented Functions / Objects	Animation / Interaction Work	Contribution (%)
1	Afzal Kabir Mahi	24-56130-1	Scenario 3 objects are Titanic ship, iceberg, stars, sea and main.cpp connection code display(), keyboard(), update(), main()), night sky	Ship collision, sinking, and rotation animation, scenario switching, and the main animation loop	34%
2	Naimul Hasan	24-55991-1	Scenario 2 objects are fish, shark, jellyfish, starfish, surface ship, sunken ship, grass, rocks, bubbles, water background	Underwater animation: fish, shark movement, anchor lift, bubble rise, grass sway	33%
3	Deawree Sayanton	23-54363-3	Scenario 1 objects are train, boat, clouds, birds, houses, trees, sky, mountains, river	Rural scene design and layout, train, boat, cloud, bird movement	33%

9. Conclusion

The project successfully shows the way in which a number of separate 2D animated scenarios can be developed and then combined into one interactive OpenGL application. It illustrates key computer graphics concepts such as coordinate transformations, procedural and timer-driven animation, alpha blending and interaction based on the keyboard.

Among its key achievements are the smooth transitions between three visually different environments and realistic environmental effects such as waving grass, floating bubbles, a rotating and sinking ship constructed entirely out of simple shapes.

The limitations consist of the use of deprecated immediate-mode OpenGL, the lack of texture mapping and audio, a fixed window resolution and the use of simplified physics approximations instead of true collision detection.

Future improvements could include porting to modern OpenGL with shaders and VBOs, adding textures and sound, introducing a basic collision engine, mouse interaction, camera movement and particle effects for smoke and splashes